Table 1: Performance concerning TPR@FPR-0.5%/100 under ChatGPT text training.

Dataset	Method	ChatGPT	GPT4All	ChatGPT-turbo	ChatGLM	Dolly	Claude	avg
Essay	DetectGPT	$37.04_{\pm 10.21}$	$24.75_{\pm 10.03}$	$5.52_{\pm 1.93}$	$21.45_{\pm 14.25}$	$15.45_{\pm 7.05}$	$4.48_{\pm 2.57}$	18.12
	Fast-DetecGPT	$9.92_{\pm 11.83}$	$37.38_{\pm 23.92}$	$51.52_{\pm 24.00}$	$15.66_{\pm 24.11}$	$0.09_{\pm 0.17}$	$0.96_{\pm 1.55}$	19.25
Reuters	DetectGPT	$4.40_{\pm 2.62}$	$0.64_{\pm 0.54}$	$2.32_{\pm 1.87}$	$2.56_{\pm 2.80}$	$0.48_{\pm 0.47}$	$3.04_{\pm 1.61}$	2.24
	Fast-DetecGPT	$72.00_{\pm 5.47}$	$9.28_{\pm 2.44}$	$94.08_{\pm 0.82}$	$93.04_{\pm 2.10}$	$1.20_{\pm 0.91}$	$9.04_{\pm 3.23}$	46.44
SQuAD1	DetectGPT	$0.35_{\pm 0.69}$	$0.36_{\pm 0.48}$	$0.47_{\pm 0.44}$	$0.58_{\pm 0.73}$	$0.58_{\pm 0.37}$	$0.81_{\pm 0.28}$	0.52
	Fast-DetecGPT	$2.66_{\pm 2.12}$	$3.45_{\pm 1.97}$	$0.94_{\pm 0.47}$	$1.73_{\pm 1.21}$	$2.67_{\pm 1.14}$	$18.27_{\pm 7.39}$	4.95

Table 2: Performance concerning AUROC/100 under ChatGPT text training.

Dataset	Method	ChatGPT	GPT4All	ChatGPT-turbo	ChatGLM	Dolly	Claude	avg
Essay	DetectGPT	$96.86_{\pm 0.97}$	$95.64_{\pm 0.45}$	$44.40_{\pm 1.73}$	$95.60_{\pm 0.62}$	$92.55_{\pm 0.58}$	$46.36_{\pm 0.71}$	78.57
	Fast-DetecGPT	$96.41_{\pm 0.96}$	$94.82_{\pm 1.19}$	$99.20_{\pm 0.22}$	$97.91_{\pm 0.49}$	$84.16_{\pm 1.97}$	$89.19_{\pm 0.76}$	93.61
Reuters	DetectGPT	$92.78_{\pm 1.34}$	$85.71_{\pm 2.02}$	$49.24_{\pm 1.65}$	$91.59_{\pm 1.67}$	$83.41_{\pm 1.73}$	$66.59_{\pm 2.53}$	78.22
	Fast-DetecGPT	$97.70_{\pm 0.16}$	$70.76_{\pm 1.59}$	$99.11_{\pm 0.12}$	$99.05_{\pm 0.20}$	$51.26_{\pm 1.29}$	$83.49_{\pm 1.70}$	83.56
SQuAD1	DetectGPT	$52.60_{\pm 2.74}$	$47.47_{\pm 1.42}$	$52.05_{\pm 1.87}$	$50.88_{\pm 2.47}$	$50.24_{\pm 2.20}$	$51.12_{\pm 2.44}$	50.73
	Fast-DetecGPT	$65.04_{\pm 0.10}$	$63.47_{\pm 1.23}$	$62.18_{\pm 0.40}$	$61.24_{\pm 0.61}$	$63.64_{\pm 0.78}$	$77.74_{\pm 0.55}$	65.55

Table 3: Performance (TPR@FPR-0.5%/100) under Google-PaLM text training in DetectRL dataset.

Method	Google-PaLM	Claude-instant	ChatGPT	Llama-2-70b	Avg.
ChatGPT-D	$87.37_{\pm 8.94}$	$67.58_{\pm 11.18}$	$69.64_{\pm 13.03}$	$99.12_{\pm 0.68}$	80.93
ChatGPT-STK	$90.11_{\pm 4.85}$	$71.73_{\pm 7.41}$	$73.56_{\pm 12.67}$	$99.21_{\pm 0.51}$	83.65
OpenAI-D	$99.00_{\pm 0.26}$	$68.59_{\pm 2.56}$	$98.94_{\pm 0.64}$	$99.91_{\pm 0.09}$	91.61
OpenAI-STK	$98.98_{\pm 0.28}$	$70.96_{\pm 3.15}$	$99.08_{\pm 0.66}$	$99.91_{\pm 0.09}$	92.23
MPU	$96.39_{\pm 1.43}$	$86.51_{\pm 1.70}$	$99.38_{\pm 0.73}$	$99.87_{\pm 0.08}$	95.54
MPU-STK	$96.41_{\pm 1.33}$	$88.41_{\pm 0.85}$	$99.41_{\pm 0.59}$	$99.91_{\pm 0.08}$	96.03

Table 4: Performance (AUROC/100) under Google-PaLM text training in DetectRL dataset.

Method	Google-PaLM	Claude-instant	ChatGPT	Llama-2-70b	Avg.
ChatGPT-D	$99.06_{\pm 0.43}$	$96.72_{\pm 1.39}$	$99.31_{\pm 0.42}$	$99.94_{\pm 0.04}$	98.76
ChatGPT-STK	$99.18_{\pm 0.32}$	$96.89_{\pm 0.92}$	$99.48_{\pm 0.24}$	$99.94_{\pm 0.04}$	98.87
OpenAI-D	$99.91_{\pm 0.02}$	$95.57_{\pm 0.63}$	$99.79_{\pm 0.05}$	$100.00_{\pm 0.01}$	98.82
OpenAI-STK	$99.91_{\pm 0.02}$	$96.20_{\pm 0.76}$	$99.81_{\pm 0.08}$	$100.00_{\pm 0.01}$	98.98
MPU	$99.81_{\pm 0.08}$	$99.24_{\pm 0.11}$	$99.94_{\pm 0.04}$	$100.00_{\pm 0.00}$	99.74
MPU-STK	$99.80_{\pm 0.07}$	$99.32_{\pm 0.11}$	$99.94_{\pm 0.04}$	$100.00_{\pm 0.00}$	99.77

Table 5: Performance against paraphrase attack Polishing under Google-PaLM text training.

Method	Google-PaLM	Claude-instant	ChatGPT	Llama-2-70b	avg.
ChatGPT-D	$85.73_{\pm 7.41}$	$70.11_{\pm 10.93}$	$94.90_{\pm 3.18}$	$97.66_{\pm 1.74}$	87.10
ChatGPT-STK	$87.95_{\pm 7.85}$	$76.33_{\pm 15.11}$	$92.29_{\pm 8.72}$	$97.25_{\pm 2.94}$	88.45
OpenAI-D	$99.70_{\pm 0.18}$	$89.59_{\pm 1.95}$	$99.98_{\pm 0.03}$	$100.00_{\pm 0.00}$	97.32
OpenAI-STK	$99.73_{\pm 0.19}$	$89.92_{\pm 1.64}$	$99.98_{\pm 0.03}$	$100.00_{\pm 0.00}$	97.41
MPU	$98.60_{\pm 0.27}$	$95.26_{\pm 1.10}$	$99.97_{\pm 0.04}$	$100.00_{\pm 0.00}$	98.46
MPU-STK	$98.52_{\pm 0.21}$	$95.46_{\pm 1.13}$	$99.97_{\pm 0.04}$	$100.00_{\pm 0.00}$	98.49

Table 6: Performance against paraphrase attack Back-translation under Google-PaLM text training.

Method	Google-PaLM	Claude-instant	ChatGPT	Llama-2-70b	Avg.
ChatGPT-D	$87.88_{\pm 6.59}$	$85.35_{\pm 6.26}$	$95.90_{\pm 3.39}$	$97.21_{\pm 2.21}$	91.58
ChatGPT-STK	$90.06_{\pm 2.35}$	$87.22_{\pm 2.71}$	$96.67_{\pm 2.03}$	$97.85_{\pm 1.03}$	92.95
OpenAI-D	$99.64_{\pm 0.08}$	$99.58_{\pm 0.18}$	$99.48_{\pm 0.15}$	$99.40_{\pm 0.29}$	99.52
OpenAI-STK	$99.62_{\pm 0.13}$	$99.59_{\pm 0.17}$	$99.50_{\pm 0.21}$	$99.40_{\pm 0.33}$	99.53
MPU	$95.90_{\pm 0.71}$	$94.25_{\pm 1.19}$	$99.78_{\pm 0.12}$	$99.78_{\pm 0.13}$	97.43
MPU-STK	$95.97_{\pm 0.79}$	$94.35_{\pm 1.25}$	$99.78_{\pm 0.12}$	$99.78_{\pm 0.13}$	97.47

Table 7: Cross-domain Performance concerning TPR@FPR-0.5%/100 under ChatGPT text training.

Method	ChatGPT	GPT4All	ChatGPT-turbo	ChatGLM	Dolly	Claude	Avg.
ChatGPT-D	$90.64_{\pm 3.07}$	$83.84_{\pm 2.97}$	$84.40_{\pm 9.06}$	$97.28_{\pm 0.64}$	$39.76_{\pm 5.36}$	$2.48_{\pm 1.48}$	66.40
ChatGPT-STK	$94.48_{\pm 1.80}$	$86.08_{\pm 1.22}$	$87.52_{\pm 4.90}$	$98.00_{\pm 0.84}$	$45.28_{\pm 4.75}$	$10.00_{\pm 3.23}$	70.23
OpenAI-D	$79.52_{\pm 35.36}$	$68.40_{\pm 20.47}$	$78.96_{\pm 33.48}$	$83.60_{\pm 28.81}$	$37.52_{\pm 11.76}$	$9.84_{\pm 2.30}$	59.64
OpenAI-STK	$80.80_{\pm 23.55}$	$66.40_{\pm 17.49}$	$82.08_{\pm 21.02}$	$88.80_{\pm 13.98}$	$33.76_{\pm 11.95}$	$16.16_{\pm 3.87}$	61.33
MPU	$100.00_{\pm 0.00}$	$87.92_{\pm 1.37}$	$89.84_{\pm 4.19}$	$99.28_{\pm 0.30}$	$55.84_{\pm 3.28}$	$85.52_{\pm 3.31}$	86.40
MPU-STK	$100.00_{\pm 0.00}$	$97.28_{\pm 1.74}$	$99.36_{\pm 1.28}$	$99.52_{\pm 0.59}$	$74.24_{\pm 12.40}$	$82.64_{\pm 15.34}$	92.17

Table 8: Cross-domain Performance concerning AUROC%/100 under ChatGPT text training.

Method	ChatGPT	GPT4All	ChatGPT-turbo	ChatGLM	Dolly	Claude	Avg.
ChatGPT-D	$98.83_{\pm 0.60}$	$97.39_{\pm 1.11}$	$97.67_{\pm 1.36}$	$99.42_{\pm 0.22}$	$80.18_{\pm 1.55}$	$42.99_{\pm 7.00}$	86.08
ChatGPT-STK	$99.27_{\pm 0.59}$	$97.56_{\pm 1.41}$	$98.07_{\pm 0.69}$	$99.39_{\pm 0.27}$	$84.41_{\pm 4.53}$	$63.22_{\pm 12.01}$	90.32
OpenAI-D	$99.46_{\pm 0.67}$	$99.03_{\pm 0.68}$	$99.44_{\pm 0.57}$	$99.52_{\pm 0.56}$	$96.15_{\pm 0.81}$	$82.63_{\pm 1.73}$	96.04
OpenAI-STK	$99.54_{\pm 0.25}$	$98.93_{\pm 0.51}$	$99.43_{\pm 0.27}$	$99.45_{\pm 0.38}$	$95.38_{\pm 0.96}$	$86.72_{\pm 2.16}$	96.57
MPU	$100.00_{\pm 0.00}$	$99.39_{\pm 0.09}$	$99.60_{\pm 0.21}$	$99.97_{\pm 0.02}$	$95.97_{\pm 0.92}$	$99.48_{\pm 0.12}$	99.07
MPU-STK	$100.00_{\pm 0.00}$	$99.78_{\pm 0.10}$	$99.91_{\pm 0.19}$	$99.99_{\pm 0.01}$	$97.94_{\pm 1.06}$	$99.34_{\pm 0.44}$	99.49